

Blinkit Replenishment Engine

The Cradle Box

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Blinkit Replenishment Engine

How We Decide What to Ship, and Where

The One Question We're Answering

“How many units of each SKU should I ship to each Blinkit warehouse this week?”

We answer this every time we run the replenishment plan. The output is a shipping manifest: warehouse by warehouse, SKU by SKU, with an invoice value check so we never ship a trickle.

How Blinkit's Supply Chain Works

Before the math, the geography matters.

```
YOUR OFFICE / 3PL
  |
  | (you ship here)
  v
WAREHOUSE (WH)      ← The Cradle Box ships to this level
e.g. Bengaluru B3
  |
  | (Blinkit distributes automatically)
  v
DARK STORES (DS)    ← Where customers actually order from
e.g. ES25 Indiranagar
e.g. ES31 Koramangala
e.g. ES41 HSR Layout
(10-25 dark stores per WH)
```

You never ship directly to dark stores. You ship to the WH. Blinkit moves stock from WH → DS on their own cadence. Your job is to keep the WH stocked.

The 7 Warehouses We Plan For

WH Code	WH Name	Key Cities
BLK_WH_1873	Bengaluru B3	Bengaluru, Hosur, Kurnool
BLK_WH_5397	Bengaluru B5	South India: Kochi, Coimbatore, Mysore, Mangaluru, Trivandrum
BLK_WH_2681	Coimbatore C1	Coimbatore
BLK_WH_5096	Faridabad	Faridabad, NCR south
BLK_WH_2010	Kundli	NCR north, Haryana
BLK_WH_2576	Noida N1	Noida, UP
BLK_WH_3201	Hyderabad H3	Hyderabad, AP, Telangana

Step 1 — Measure Daily Sales Rate (ADS)

ADS = **Average Daily Sales per Dark Store**, computed over the most recent 30 days.

We pull this from Blinkit's daily performance report (the CSV you download each morning and the scraper saves automatically).

THE RAW FORMULA

$$\text{raw_ADS} = \frac{\text{orders on available days}}{\text{count of available days}}$$

"Available day" = a day when Column Q in the performance CSV showed inventory was present at that dark store.

We only divide by days the SKU was actually in stock. If a DS was OOS for 20 of 30 days, we divide by 10, not 30 — because we're measuring selling velocity, not penalising OOS.

BUT WHAT ABOUT OOS DAYS?

If a DS ran out of stock for 10 days, those days had zero recorded orders — but demand didn't disappear. Customers just couldn't buy.

We **impute** missing demand for OOS days:

$$\text{imputed_ADS} = \frac{(\text{raw_ADS} \times \text{avail_days}) + (\text{P75} \times \text{oos_days})}{30}$$

P75 is the 75th percentile ADS across all reliable dark stores for that SKU in the same city. If no city data exists (e.g. a brand-new city), it falls back to the global P75 across all cities.

Why P75, not average? It's mildly bullish — assumes OOS days had at least median+ demand. Better to ship slightly more than to go OOS.

RELIABILITY GATE

A dark store must have **at least 5 available days** in the 30-day window to be counted as a reliable data point for P75 computation. Stores with fewer days are imputed using the city/global P75 but don't influence it.

Step 2 — Aggregate to WH Level

For each WH × SKU pair:

$$\text{total_ADS} = \text{SUM of imputed_ADS across all active, eligible DSes under that WH}$$

Eligible means the dark store has status = `active` in our eligibility table. We exclude:

Status	What It Means	Included?
active	DS selling the SKU	Yes
launch_awaited	City not yet activated in Blinkit panel	No
darkstore_closed	Physical store shut	No
sku_moved_out_low_sales	SKU redistributed out by Blinkit	No
sku_city_exited	SKU pulled from city entirely	No
ds_choked	DS throttled by Blinkit	Counted but flagged

Step 3 — Compute Target Stock

```

target_stock = total_ADS × (coverage_days + transit_buffer)
              = total_ADS × (30 + 7)
              = total_ADS × 37

```

Parameter	Value	Reasoning
Coverage days	30	Monthly shipping cadence — one shipment covers one month
Transit buffer	7	In-transit safety stock — stock is en route and not yet in WH

THE FLOOR RULE

If a SKU is newly launched or has been OOS so long that its computed ADS is near zero, the formula would give a dangerously low target stock — potentially zero, meaning we'd never seed the WH.

Floor: `target_stock = max(computed_target, active_DS_count)`

This guarantees at least 1 unit per active dark store, so live stores never starve.

Step 4 — Subtract Effective Stock

We don't ship what we already have:

```
effective_stock = units_wh + units_incoming + units_transit + units_ds
```

Component	What It Is
<code>units_wh</code>	On WH shelf right now
<code>units_incoming</code>	Dispatched from your location, not yet received
<code>units_transit</code>	In WH → DS pipeline
<code>units_ds</code>	Already on DS shelves

```
units_to_ship = max(0, target_stock - effective_stock)
```

Step 5 — The ₹1.5 Lakh Gate

We only actually ship to a WH if the total invoice value of the shipment crosses **₹1,50,000**.

Below this threshold: `DEFER` — accumulate more SKUs or wait until the next cycle.

This prevents sending small parcels that cost more in logistics than the inventory value justifies.

The Full Flow, End to End

```

Daily performance CSV
  |
  v
blinkit_performance_detail (DB)
  |
  v
compute_ads()
├ Per DS: raw_ADS on available days
├ Impute OOS days with city P75 (or global fallback)
└ ADS = (raw_orders + imputed_orders) / 30
  |
  v
Filter to active, eligible DSes
Group by WH × SKU → total_ADS
  |
  v
target_stock = total_ADS × 37
(floored at active_DS_count)
  |
  v
Subtract effective_stock from inventory snapshot
  |
  v
units_to_ship per WH × SKU
  |
  v
Apply ₹1.5L gate per WH
  |
  v
Replenishment Plan Excel
├ Sheet: Ship Now (WHs that pass the gate)
├ Sheet: Full Plan (all WH × SKU with target, stock, gap)
├ Sheet: Overview-WH
└ Sheet: Summary

```

A Worked Example

SKU: TCB001 (Tiny Splash Pink) **WH:** Bengaluru B3

Dark Store	Avail Days	OOS Days	Orders (avail days)	raw_ADS
ES25 Indiranagar	18	0	4	0.222
ES31 Koramangala	15	5	3	0.200
ES41 HSR Layout	12	8	1	0.083
ES58 Whitefield	0	9	0	— (imputed)

City P75 for TCB001 in Bengaluru = **0.18**

Dark Store	Imputed ADS
ES25	$(0.222 \times 18 + 0.18 \times 0) / 30 = \mathbf{0.133}$
ES31	$(0.200 \times 15 + 0.18 \times 5) / 30 = \mathbf{0.130}$
ES41	$(0.083 \times 12 + 0.18 \times 8) / 30 = \mathbf{0.081}$
ES58	$(0 \times 0 + 0.18 \times 9) / 30 = \mathbf{0.054}$

Total ADS = 0.133 + 0.130 + 0.081 + 0.054 = 0.398

Target stock = $0.398 \times 37 = 14.7 \rightarrow 15$ units
 Effective stock = 3 (WH) + 0 (incoming) + 0 (transit) + 1 (DS) = 4
 Units to ship = $15 - 4 = 11$ units

What Can Go Wrong

Issue	What You See	What It Means
All Kundli DSes are OOS for a new SKU	ADS driven entirely by global P75 fallback	The OOS imputation is using South India demand as proxy for Delhi — treat with caution
WH-OOS blind spot	total_ADS = 0 despite SKU being listed	WH has been dry all period, so no performance data accumulated. Floor kicks in (target = DS count), but it's a guess
DS not launched	<code>launch_awaited</code> status	City activation not done in Blinkit panel. Do NOT ship stock — it won't reach dark stores
DS choked	<code>ds_choked</code> status	Blinkit has throttled that DS. Counted in overview but excluded from ADS

Key Numbers

Parameter	Value
Lookback window	30 days (rolling from latest data date)
Coverage target	30 days
Transit buffer	7 days
Min available days for reliable ADS	5 days
WH shipment gate	₹1,50,000 per WH
OOS imputation	city P75 → global P75 fallback

Generated by The Cradle Box demand planning system · [tcb/replenishment.py](#)